

TASHKENT INTERNATIONAL UNIVERSITY · FACULTY OF
ECONOMICS Point and interval assessment

Week 9 · Lectures 1 and 2

Tashkent International University · Faculty of Economics

2025/2026 academic
year



It's time to keep the promise

μ was also given on the midterm exam. Someone would give you μ for every problem.

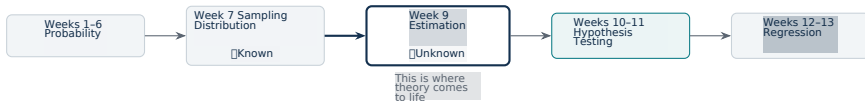
Audience question But μ is a parameter of the prime set. If we knew it, we would also know σ . And then we wouldn't need statistics.

So μ is never known.

This week's two questions 1. When you evaluate a parameter with a single number — what number is it? And how good is it?

2. What goes wrong when you replace μ with $\bar{x}$ and how do we fix it?

Theorem Answer: One number is not enough. What is needed is an interval, and it carries its own uncertainty.



Learning Objectives for Week 9

By the end of this week, you will:

1. Describe and measure three properties of a Point Estimator—immutability, efficiency, and robustness.
2. Demonstrate with a simulation why a Student's t -distribution is needed instead of a z when σ is replaced by s .
3. Construct a Confidence Interval for the mean and interpret it correctly.
4. Construct an interval for the proportion and identify where the Wald formula breaks down.
5. Construct a bootstrap-free interval.
6. What does the interval not cover?

How close is this to μ — and how can we tell without knowing μ ?

CLO6 — Construct and interpret confidence intervals for means and proportions. Bloom's: Application / Analysis.

This week's method: We know the original set ($n = 35.8050$, $n = 37.4048$ — 2,467 movies). So we don't say the error of each method, we measure it.

Point estimator

Estimator is a formula, score is a number

Estimate is the number that comes out after applying this rule to the sample in your hand: $\bar{x} = 35.84$.

A common mistake Properties (immutability, etc.) belong to the estimator, not to the score. “My score is immutable” is a meaningless sentence: a single number cannot be immutable.

Three properties

Immutability: $E[\hat{\theta}] = \theta$ — on average, it hits the target.

Efficiency: $\text{Var}(\hat{\theta})$ is small — less spread around the target.

Validity: $\lim_{n \rightarrow \infty} \text{Var}(\hat{\theta}) = 0$ — the error disappears as the information increases.

Audience question The three are different things. It is possible to be unshifted and very scattered. It is also possible to be shifted and more precise — we will see that now.

Non-movement — measured, not stated

Main set: 2,467 film budgets. $\mu = 35.8050$, $\sigma^2 = 1399.12$, median = 22.00. 20,000 samples for each α

Estimator	$\alpha=5$	$\alpha=10$	$\alpha=30$	Target
$\bar{X}$	35,674	35,601	35,814	$\mu = 35,805$
$s^2 = \frac{\sum(x-\bar{x})^2}{n-1}$	1389,6	1383,9	1400,1	$\sigma^2 = 1399,1$
α	1111.6	1245.5	1353.5	$\alpha=1399.1$
sample median	24.75	23.20	22.00	26.55

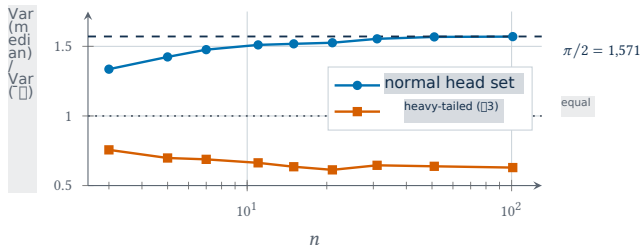
Theorem The differences are of the order of Monte Carlo noise (± 18.7 for α at $\alpha=5$). At $\alpha=10$ they slightly exceed the limit — but the sign changes in the other five seeds ($-0.20 + 0.04 + 0.05 - 0.09 - 0.06$): no standard deviation.

uchun $\pm 18,7$). $\alpha=10$ da ular chegaradan biroz oshadi — ammo boshqa beshta urug'da ishora o'zgaradi ($-0,20 + 0,04 + 0,05 - 0,09 - 0,06$): muntazam og'ish yo'q.

Common mistake

The shift of α is not noise: -287.5 at $\alpha=5$, theoretical $-\alpha/\alpha = -279.8$ — and it is negative at every seed. Regular, predictable, correctable.

Efficiency: there is no such thing as a “best estimator”



Theorem In normal, the ratio tends to $\pi/2$: the median is 57% more spread out than the mean — the mean wins.

Common mistake

In heavy tail, the ratio is ≈ 0.63 : the median is more accurate — now the median wins.

Audience question

Conclusion — and it is the idea of the whole course. There is no universal answer to the question “Which estimator is better?” — the answer depends on the shape of the head set.

Shifted — but more accurate

Mean Square Error (MSE)

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = (E[\hat{\theta}] - \theta)^2 + \text{Var}(\hat{\theta})$$

dispersion

Normal prime set, $\theta = 1$ we evaluate:

θ	shift(θ)	MSE(θ)	shift(θ)	MSE(θ)
3	+0,007	1,026	-0,329	0,564
5	-0,002	0,502	-0,201	0,362
10	+0,005	0,227	-0,095	0,193
30	-0,005	0,069	-0,038	0,066

Unexpected result

θ is shifted, but its MSE is smaller at each θ . That is, it is closer to the true θ on average.

Theorem

Theory confirms: $MSE(\theta) = 2\theta - 1$,

$$MSE(\theta) = 2\theta - 1$$

$\theta = 3$: 1.000 and 0.556; simulation 1.026 and 0.564.

Audience question

Why $\theta - 1$ anyway? Because the non-shifting carries over to other formulas, while the MSE advantage does not. This is a compromise, not a divine truth.

□-distribution

What happens if we change n to ∞ ?

Week 7

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

The denominator is a fixed number. Only the sum is random.

Now

$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ The denominator is also random! The ratio of two random quantities. If n is small, Z explodes.

Result: Z is not normal — the tails are heavier.

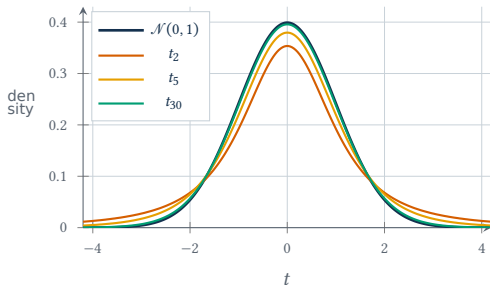
From a normal population, 40,000 samples:

n	df	SD($\bar{X}$)	SD(Z)	$P(Z > 1.96)$
3	2	1,003	4,612	0,1905
5	4	1,000	1,403	0,1184
10	9	1,005	1,137	0,0821
30	29	0,993	1,028	0,0579
100	99	1,001	1,012	0,0533

Audience question

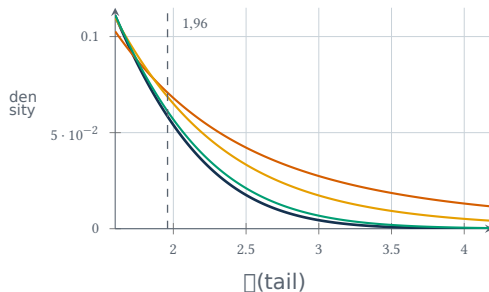
$P(|Z| > 1.96)$ is never 0.05. At $n=5$ it is 0.118 — that is, the “95%” interval based on 1.96 is actually 88%. SD(Z) corresponds to the theoretical $\sqrt{df/(df-2)}$; at $df=2$ it is infinite.

χ^2 -distribution: same center, heavier tails



Theorem The four curves on the left almost overlap — there is no difference in the center. The difference is only in the tail. And we use the tail: the 95% limit lies in the tail.

Let's zoom in on the tail



Audience question

Single parameter: Degrees of Freedom $df = \chi^2 - 1$. Why $\chi^2 - 1$?
Because χ^2 was used to calculate χ^2 — one degree of freedom was “spent” (this is exactly the answer from week 7).

Critical values of t^* — what is the cost?

df t^* 0.975 greater than 1.96

1	2	12,706	+548 %
2	3	4,303	+120 %
4	5	2,776	+42 %
9	10	2,262	+15 %
19	20	2,093	+7 %
29	30	2,045	+4 %
99	100	1,984	+1 %
∞	—	1,960	0 %

How to read this

Using t^* widens the range. This is not a penalty, it is honesty.

Our lack of knowledge of t^* introduces additional uncertainty; t^* adds this uncertainty to the range.

Audience question Rule of thumb: At $df \geq 30$, the difference is less than 4% — that's why most textbooks say "you can use t^* in a large sample."

But why do you use it? The calculator calculates both equally easily. Always t^* .

**Confidence
interval**

Confidence interval for the mean

$\bar{x}$ -interval

$$\bar{x} \pm t_{1-\alpha/2}(n-1) \cdot \frac{\sqrt{\hat{\sigma}^2}}{\text{precision}} \text{ i.e. } \text{estimate} \pm \text{confidence} \times \text{SE}$$

Actual data: Uzbekistan wheat yield, 2009–2018 ($n = 10$), t/ha.
Source: OWID/FAOSTAT.

$$\bar{x} = 4.6350 \quad \hat{\sigma} = 0.2446 \quad \text{SE} = 0.2446/\sqrt{10} = 0.0774$$

Confidence interval $\bar{x}$ -value Interval width			
90 %	1,833	[4,4932; 4,7768]	0,2836
95 %	2,262	[4,4600; 4,8100]	0,3500
99 %	3,250	[4,3836; 4,8864]	0,5028

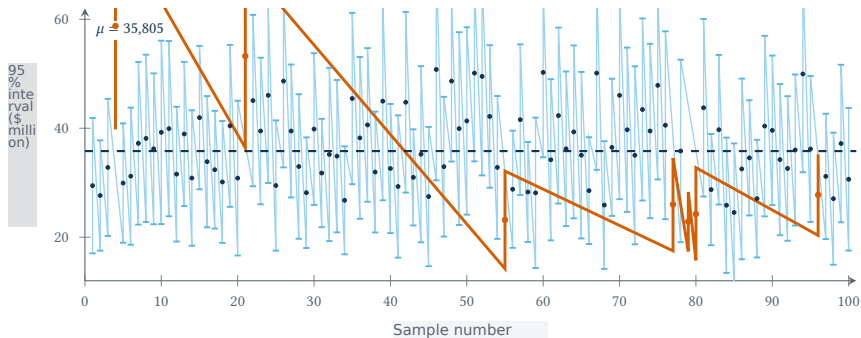
Three-step rule

1. Check the conditions: random sampling; if n is small, the population is approximately normal.
2. Find $\bar{x}$, $\hat{\sigma}$, SE, and $1 - \alpha/2(n-1)$.
3. Write the result in context—not as a number, but as a sentence.

Common Error

The 99% interval is 1.437 times wider than the 95%. More confidence doesn't come for free: its price is precision.

What does the interval mean: 100 samples

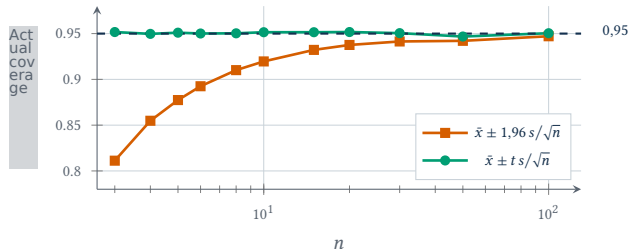


Theorem $\square = 30$, 100 samples. 93 covered $\square$, 7 did not (red). Expected 95 — the difference is a random variation.

Audience question

Note: the intervals are different ENDA. The narrowest is 10.91, the widest is 40.73 — 3.7 times. Reason: $\square$ is different in each sample. If $\square$ were known, they would all be the same ENDA.

Is it really necessary? — let's count the coverage



Theorem Normal sample size, 20,000 samples. The $\square$ line is 0.95 at every $\square$ — exactly the promised coverage.

A common mistake

The 1.96 line is 0.811 at $\square = 3$: one in every 5 intervals is wrong — the so-called “95%” method.

Audience question

This graph is the whole basis of the $\square$ -distribution. You don't need to memorize it — just look at it: if you use $\square$, the critical value of $\square$ makes you overconfident.

What happens if the normality condition is violated?

Same experiment, but the sample is real (movie budget, skewness 1.62) — not normal.

n and σ and σ and a known σ -interval width

5	0,786	0,855	0,956	81,8
15	0,890	0,907	0,955	39,9
30	0,922	0,931	0,953	27,4
100	0,943	0,946	0,952	14,7
400	0,950	0,951	0,951	7,3

Compare column 2 with the normal sample

In the normal sample, the σ -interval at $n=5$ would give 0.9510; here 0.8552. The remaining 0.096 deficit is FROM THE SHAPE of the entire sample. σ does not correct for normality — it only corrects for the unknown σ .

Column 3: Skewness itself is not a fault

If σ is known, the coverage is not less than 0.9559 — 0.95. So the skewness itself does not cause harm. The harm comes from using σ in a skewed prime set: where σ and σ act together. MLT cancels this out with σ .

Why exactly $\hat{\mu}$ and skewness together are harmful

The mathematical basis of the $\hat{\mu}$ -distribution is based on one condition: $\hat{\mu}$ and $\hat{\sigma}$ must be independent. Let's measure this.

	$\hat{\mu}$ Corr($\hat{\mu}$, $\hat{\sigma}$) skewed	Corr($\hat{\mu}$, $\hat{\sigma}$) normal
5	0,793	0,003
10	0,781	-0,003
30	0,784	0,015
100	0,782	-0,017

In normal, the correlation is zero — the condition is met. The skewness is 0.78, and even with $\hat{\mu}$, it still breaks down.

The consequence — the two errors add up in one direction

If the sample does not take a large value in the tail: it will be lower than $\hat{\mu}$ and it will be smaller. The interval is both low-centered and narrow.

	$\hat{\mu}$ below	$\hat{\mu}$ above
5	97,6 %	2,4 %
10	95,4 %	4,6 %
30	86,7 %	13,3 %
100	74,4 %	25,6 %

Direction of non-covering intervals. In a normal prime set this would be 50/50.

Theorem Summary. Skewness does almost no harm to the $\hat{\mu}$ -interval ($\hat{\mu}$ known) — there is no $\hat{\mu}$. The harm is $\hat{\mu}$ and skewness together. MLT symmetricizes the error (the last column moves towards 50%), but slowly.

Incorrect Why

“There is a 95% chance that μ is in this interval.” μ is a fixed number. The interval is random. The probability for a given interval is either 0 or 1. 95% is the long-term coverage rate of the method. “95% of the observations are in this interval.” Only 20.5% of the 2,467 movies lie in the single $\mu = 30$ interval [26.4; 52.3]. The interval needed for 95% of the observations is [1.0; 150.0]. “If two intervals intersect, there is no difference.” Intersection does not mean “no difference.” The correct method is to construct an interval for the difference (week 11). “The range is wide — the data is bad.” The range is $2\sigma/\sqrt{n}$: the natural variance (σ), the investment (n), and your choice (confidence level). A wide range is an honest answer. “ μ is large — so the range is correct.” The range only covers random error. If the sample is skewed, it will be narrow around the wrong place. (Week 7: Literary Digest.)

Space for sharing

Proportion: standard formula — and its flaw

Wald interval (in textbooks)

$$\hat{p} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Condition: $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$.

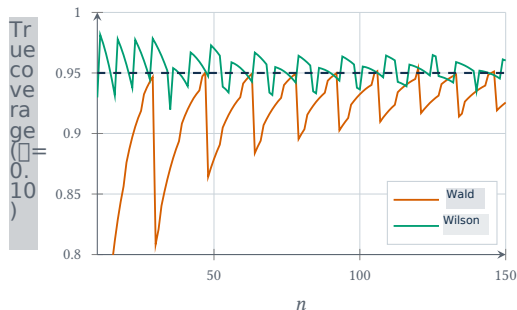
Example: $\pi = 0.34$ out of 100 movies are PG-13. $\hat{\pi} = 0.34$; $34 \geq 10$, $66 \geq 10$

SE = $\sqrt{0.34 \cdot 0.66/100} = 0.0474$

Wald: [0.2472; 0.4328] width 0.1857

Wilson: [0.2546; 0.4372] width 0.1826

True $\pi = 0.3632$ — both covered.



Common mistake

Coverage fluctuates — because the number of successes is discrete. Don't make conclusions based on a single π . Average over $\pi = 10 \dots 200$ ($\alpha = 0.05$): Wald 0.874, Wilson 0.955.

Why Wald breaks down — and what to do

Exact coverage (not simulation — sum over all π), averaged over $n = 10 \dots 200$:

	Wald	Wilson	Agresti-Coull
0,05	0,874	0,955	0,967
0,10	0,918	0,953	0,959
0,30	0,940	0,951	0,952
0,50	0,943	0,950	0,951

Reason. We put $\hat{\pi}$ in its SE. If by chance $\hat{\pi}$ turns out to be small, the SE turns out to be small too — the range is reduced by half. It is fatal at the extreme values: if $\hat{\pi} = 0$, the range is $[0; 0]$.

Agresti-Coull — the simplest fix

Exact form: $\tilde{\pi} = \frac{\pi + \pi^2/2}{\pi + \pi^2}$, then Wald.

At $\pi = 1.96$, $\pi^2/2 = 1.92 \approx 2$, so in practice “add 2 successes and 2 failures”

”: $\tilde{\pi} = \frac{\pi + 2}{n + 4}$

$\pi = 100$, $\pi = 34$: the exact form $[0.2544; 0.4374]$, the $+2/+4$ form $[0.2547; 0.4376]$ — the difference is insignificant.

Audience question

Why do we study Wald anyway? Because it is everywhere: in textbooks, reports, software output. You need to know it and know where it breaks down. In practice: π small or π close to the edge — Wilson (IMS 2e recommendation).

Bootstrap

Bootstrap: what if there is no formula?

For the problem σ , we have the formula $SE = \sigma/\sqrt{n}$.

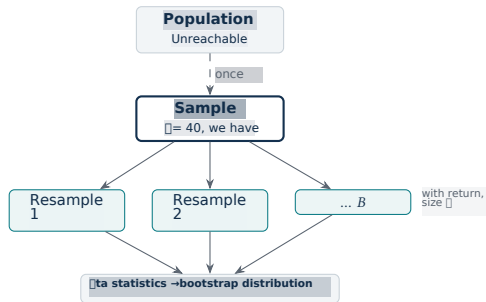
What about the median? For the quartile? For the coefficient of variation? For the ratio of two medians?

Most don't have a formula — or they do, but they're terrible.

Efron's (1979) idea

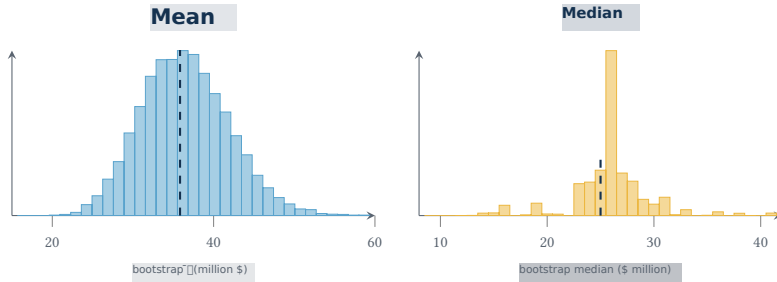
We can't resample from the original set.

But we can resample by returning from the sample itself — because the sample is the best available representation of the original set.



95% interval — 2.5 and 97.5 percentiles of the bootstrap distribution. No formula at all.

Bootstrap in progress — true sample, $\mu = 40$



$\bar{x}$ -range [24.67; 47.00]
Bootstrap [25.95; 47.22]

Bootstrap validates formula.

Bootstrap range [15.00; 32.50] The true population median is 22.00 — covered

This is the basic value of bootstrapping.



The median histogram is notched — the median only takes the values present in the sample.
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Bootstrap is not a magic wand

□ bootstrap classic □

10	0,884	0,924
30	0,920	0,926
100	0,938	0,930

Both are below 0.95 — the set is skewed.
Bootstrap is worse at small n .

Sample maximum = 151.0. Out of 4,000 bootstraps, the maximum takes only 6 different values, and 63% of them are exactly the original maximum.

Bootstrap never gives a number greater than 151 — the interval has shifted.

Audience Question Rule: Bootstrap works for smooth statistics (mean, median, correlation). Not for extremes. It replaces the formula, not the data.

**Summar
y**

Misconception Why it's wrong

" $\bar{x}$ with 95% probability in this interval." $\bar{x}$ is not random. 95% is a property of the method. It covered 93 out of 100 intervals — here's the definition. "If $n \geq 30$, then $\bar{x}$ can be used." Yes, the difference is less than 4%. But why? Using $\bar{x}$ is equally easy and always correct. " $\bar{x}$ solves the problem of normality." No. $\bar{x}$ only solves the unknown μ . In a skewed set, $\bar{x}$ -interval covers 0.855 at $n = 5$. "The unshifted estimator is the best." $\bar{x}^2$ is shifted, but its MSE is smaller. "Good" is a question of what criterion. "The mean is always better than the median." Only in a nearly normal population. In a heavy tail, the median is $\approx 37\%$ more accurate. "Bootstrap is magic; no formula needed." Bootstrap cannot generate more information than sampling. In a skewed population, it does not yield 0.95 either. "Large n —interval is reliable." The interval covers random error. It does not cover drift and does not disappear with n .

Main results of week 9

Must remember

$$\text{MSE}(\hat{\mu}) = \text{shift}^2 + \text{Var}(\hat{\mu})$$

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

$$\text{Mean: } \hat{\mu} \pm \frac{1 - \alpha}{2} (n - 1) \frac{s}{\sqrt{n}}$$

$$\text{Proportion: } \hat{p} \pm \frac{1 - \alpha}{2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

$$\text{Size: } \left(\frac{z_{\alpha/2}}{E} \right)^2, \quad n \geq \frac{z^2}{4E^2}$$

In three sentences

1. One number is never enough — the estimate must carry its uncertainty with it.
2. Not knowing μ is not free: μ adds this cost to the width of the interval.
3. The interval only measures random error. There is no formula for shift no — just good design.

Next step

Now we can answer the question “How much is μ ?” In week 10, another question comes up: “Is μ equal to such and such a number?” — hypothesis testing, the p -value, and its misinterpretations.

Spoiler: confidence intervals answer this question too.

References

Main reading

►Cetinkaya-Rundel M., Hardin J. Introduction to Modern Statistics (IMS, 2nd ed.), Chapters 12–17 — confidence intervals and bootstrap. Open source: openintro.org/book/ims ►Newbold P., Carlson W., Thorne B. Statistics for Business and Economics (9th ed.), Chapters 7–8.

Additional and reference articles

►Efron B. (1979). Bootstrap methods: another look at the jackknife. *Annals of Statistics* 7(1), 1–26. — original article on bootstrap. ►Brown L., Cai T., DasGupta A. (2001). Interval estimation for a binomial proportion. *Statistical Science* 16(2), 101–133. — The fallacy of the Wald interval and Wilson's recommendation here. ►“Student” (Gosset W.S., 1908). The probable error of a mean. *Biometrika* 6(1), 1–25. — The birth of $\bar{x}$. ►Morey R. et al. (2016). The fallacy of placing confidence in confidence intervals. *Psychonomic Bulletin & Review* 23, 103–123. Interactive

►seeingtheory.brown.edu — “Confidence Interval”; rpsychologist.com/d3/ci — live animation of the coverage. ►Reference: [OWID/FAOSTAT \(productivity\)](#), [movies-budget.csv](#).